

MEDICAL DEVICE MARKETING
UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

2. Financial Targets

Financial targets are the **specific financial goals established for a product or project**.

They help determine whether the product is commercially viable.

Important financial targets

1. Development budget

Maximum amount allocated for product development.

2. Manufacturing cost

Cost required to manufacture one unit.

3. Selling price

Price at which the product is sold.

4. Revenue target

Expected income from product sales.

5. Profit target

Expected profit from the product.

6. Break-even point

The sales level at which total revenue equals total cost.

7. Return on Investment (ROI)

Measures the return generated from an investment.

$$\text{ROI} = (\text{Net Return} / \text{Investment}) \times 100$$

Example

Suppose:

- Manufacturing cost = ₹8,000/device
- Selling price = ₹15,000/device
- Expected sales = 1,000 devices

The company can estimate its revenue, contribution margin, operating costs, and expected profit before deciding whether to proceed with the project.

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7. Product acceptance

A product developed with appropriate clinical input is more likely to be accepted by healthcare professionals.

Methods of Obtaining Physician Input

- Interviews
- Questionnaires
- Focus groups
- Clinical observations
- Product demonstrations
- Advisory panels
- Prototype evaluation
- Clinical studies

Example

For a **smart prosthetic limb**, engineers may develop the sensors and control system, while physicians and rehabilitation professionals can provide input regarding:

- Patient comfort
- Gait requirements
- Safety
- Socket fit
- Rehabilitation needs
- Clinical usability

4. FDA Approval

Introduction

The **U.S. Food and Drug Administration (FDA)** regulates medical devices marketed in the United States. FDA's Center for Devices and Radiological Health (CDRH) regulates firms that manufacture, repack, relabel, or import medical devices for the U.S. market.

An important point for students is that **not every medical device receives an "FDA approval."** Depending on the device and regulatory pathway, the product may be **510(k) cleared, De Novo classified, PMA approved, or exempt from a premarket submission.**

Objectives of FDA Regulation

FDA regulation aims to ensure that medical devices:

- Are reasonably safe
- Are effective for their intended use

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- Corrective and preventive actions
- Complaint handling
- Risk management
- Records

Step 7 – Prepare Regulatory Submission

The manufacturer prepares the appropriate regulatory submission.

Depending on the pathway, this may include:

- Device description
- Intended use
- Indications for use
- Design information
- Risk information
- Performance data
- Test results
- Clinical data where required
- Manufacturing information
- Quality-system information
- Labeling

Step 8 – Labeling

Medical device labeling is an important regulatory requirement.

FDA's general device labeling requirements are contained in **21 CFR Part 801**.

Labeling may include:

- Device name
- Manufacturer information
- Intended use
- Indications
- Instructions for use
- Warnings
- Precautions
- Contraindications

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- Installation instructions
- Maintenance information

Labeling must not provide misleading information or unsupported claims.

Step 9 – FDA Review

FDA reviews the submitted information according to the applicable pathway.

The review may involve:

- Technical evaluation
- Clinical evaluation
- Statistical evaluation
- Manufacturing/quality considerations
- Labeling review
- Risk assessment

FDA may request additional information where necessary.

Step 10 – Regulatory Decision

Depending on the pathway, the outcome may be:

- 510(k) clearance
- De Novo classification
- PMA approval
- HDE approval
- Exemption where applicable

Step 11 – Post-Market Requirements

FDA oversight does not end when a device reaches the market.

Post-market activities may include:

- Complaint handling
- Medical device reporting
- Corrective actions
- Recalls where necessary
- Post-market surveillance
- Manufacturing quality monitoring